



江苏省苏州中学 国际部
Suzhou High School - International Division

SZZX-ID Bridge Program Physics

Friction

Class Outline

Title of Unit	Forces and Newton's Laws
Title of Lesson	friction
Starter Activity	Spot friction around us
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- Questioning: frictions in life
	- Lecturing: What does the coefficient of friction mean for static and dynamic friction
	- demonstration: static and dynamic friction
	- inquiry: what is supplying the normal force in the stacked sticky pad case?
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- Encourage students to share their opinions during class
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Understand the similarity and differences between static friction and dynamic friction 2. Understand the direction of frictional force 3. Understand coefficient of friction μ for dynamic friction and max. static friction 4. Know examples of friction in life, and measures to increase/decrease friction
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication: Using terminology, symbols and communication conventions consistently and correctly
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Thinker, communicator
Teaching Activities	Demonstration: measuring static and dynamic friction force with a sensor Video: friction (active learning, use proper terminologies) Worksheet to check the effectiveness of learning.
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Homework: draw a FBD whenever possible

Vocabulary



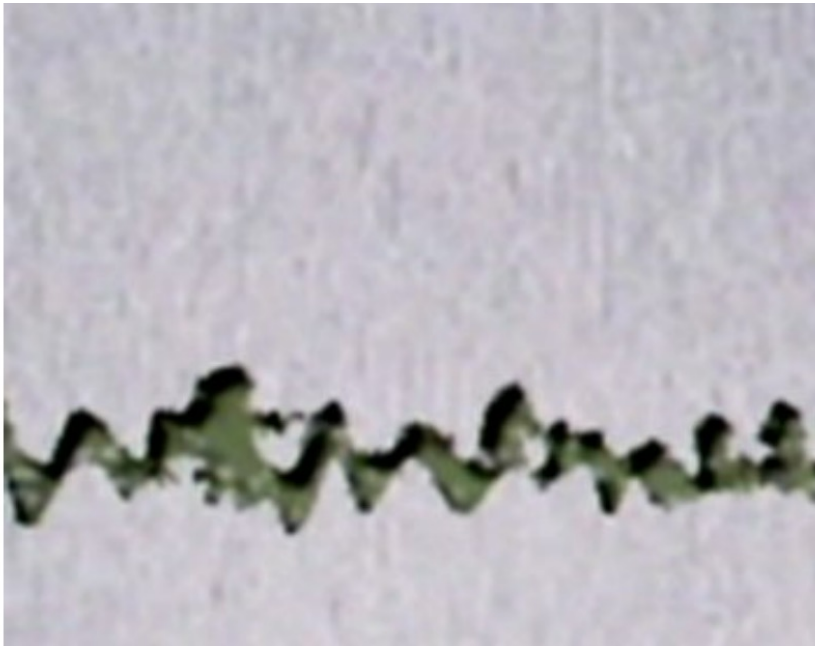
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- Static 静止的
- Dynamic 运动的
- Microscopic 微观
- Jag 锯齿状、粗糙的
- Coefficient 系数
- Normal force 支撑力
- Empirical laws 经验法则

Why friction force? An explanation



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At an interface between two objects, the surfaces may appear to be smooth, however, on a microscopic scale the surfaces are pitted and jagged. When another similar surface is in contact with it, the irregularities and peaks interlock and try to act against any relative moment

Microscopic irregularities of smooth surface in contact,
<http://www.splung.com/content/sid/2/page/friction>

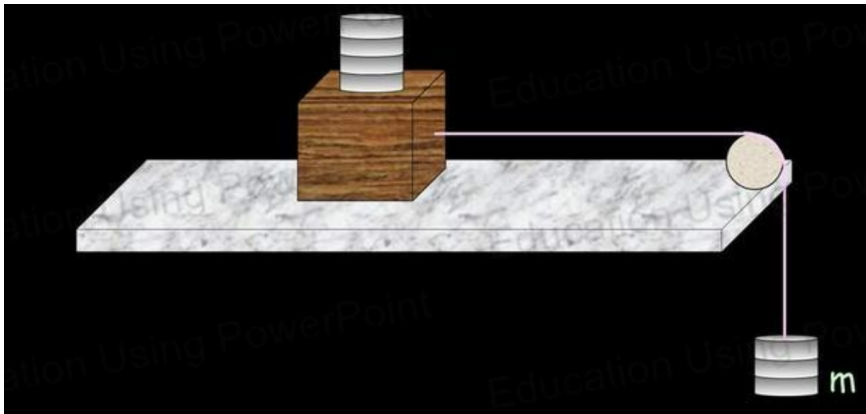
What is friction force



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Friction force

- Oppose the direction of **relative motion or tendency of motion**
- Along the direction of the contact surfaces
- A contact force
- Dynamic friction: when one body slides over another
- Static friction: when one body has a tendency to move against another.



if all objects are at rest (in equilibrium)

Qn: Is there friction force between the wooden block and the top weight?

Qn: Is there friction force between the wooden block and the table?

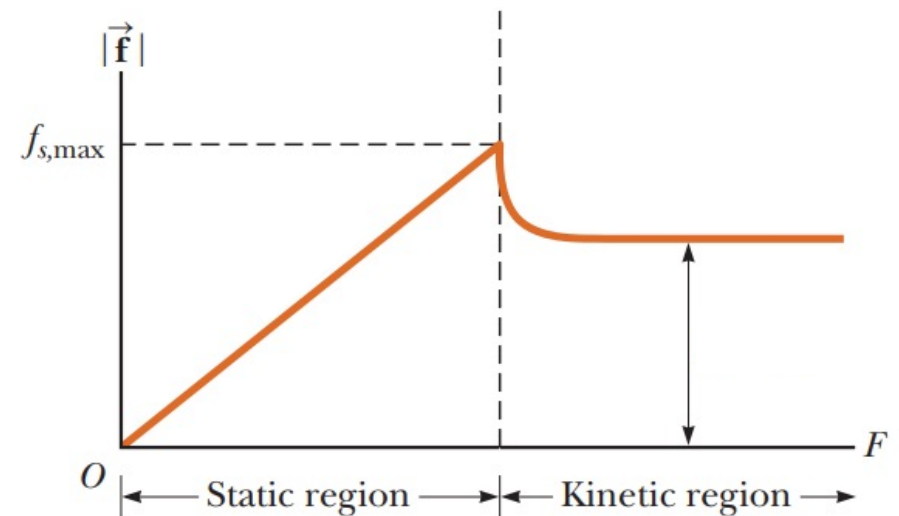
Static and Dynamic (kinetic) Friction



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What happens when you try to push a heavy box?

- At first it does not move
- When the force to push is increased, it starts to slide
- After it starts to move, it feels a little easier to push it further forward



Friction laws – (they are empirical laws)



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The force of dynamic friction is equal to:

$$f_d = \mu_d N$$

where N is the normal reaction force between the surfaces and μ_d is the coefficient of dynamic friction

The **max.** force of static friction is equal to:

$$f_{s-\max} = \mu_s N$$

(or $f_s \leq \mu_s N$)

where N is the normal reaction force between the surfaces and μ_s is the coefficient of static friction, with $\mu_s > \mu_d$

- The area of contact between the two surfaces does not affect the frictional force
- The force of dynamic friction does not depend on the speed of sliding (or the impact is small enough to be neglected)
- Coefficients of friction depend on the two surfaces and their roughness

A simple way to measure μ_s

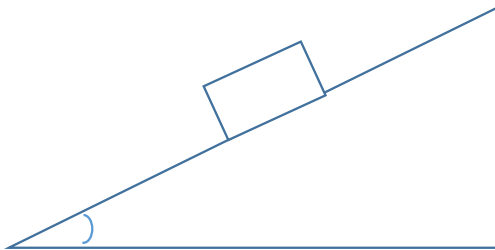


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A block stays on a slope with a small angle α .

Increase the angle slowly until the moment the block just starts to move. At that moment, the angle of the slope is α

What is the coefficient of static friction between the block and the slope?

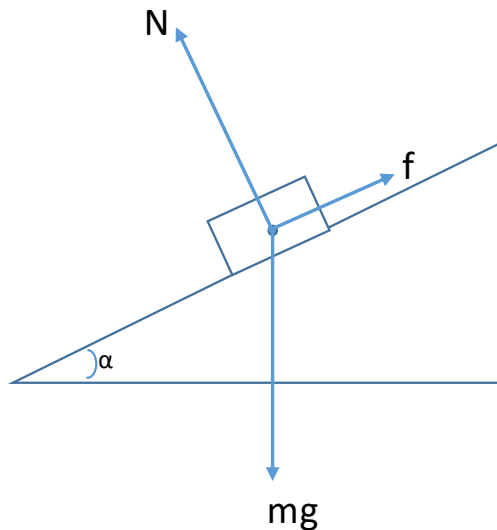


1. Draw the forces
2. Resolve forces parallel to the slope
3. Resolve forces perpendicular to the slope

A simple way to measure μ_s



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What is the coefficient of static friction between the block and the slope?

At that moment, $f_s = f_{s,\max}$

1. Draw the forces
2. Resolve forces parallel to the slope

$$f = f_{s,\max} = mg \sin \alpha$$

3. Resolve forces perpendicular to the slope

$$N = mg \cos \alpha$$

$$\text{The coefficient of static friction } \mu_s = f_{s,\max} / N = \tan \alpha$$

Think pair and share



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- When there is normal force, must there be friction?
- When there is friction, must there be a normal force?
- What can you conclude on the angle between normal force and friction force?

Example 1



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A brick of weight 50 N rests on a horizontal surface. The coefficient of static friction between the brick and the surface is 0.60 and the coefficient of dynamic friction is 0.20 . A horizontal force F is applied to the brick, its magnitude increasing uniformly from zero. Once the brick starts moving the pulling force no longer increases. Estimate the net force on the moving brick.

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The maximum frictional force that can develop between the brick and the surface is:

$$f_s = \mu_s R$$

which evaluates to:

$$0.60 \times 50 = 30 \text{ N}$$

So motion takes place when the pulling force is just barely larger than 30 N.

Once motion starts the frictional force will be equal to $\mu_d R$, i.e.

$$0.20 \times 50 = 10 \text{ N}$$

The net force on the brick in that case will be just larger than $30 - 10 = 20 \text{ N}$.

Example 2



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When two applied forces are pushing the block (as shown in the figure), the block stays at rest.

$F_1 = 10\text{N}$ and $F_2 = 2\text{N}$.

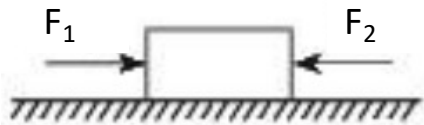
When F_1 is removed, what is the force of friction acting on the block?

- A. 10N to the left
- B. 6N to the right
- C. 2N to the right
- D. 0

Example 2



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When F_1 is removed, what is the force of friction acting on the block?

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- B. 6N to the right
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- D. 0

- When F_1 and F_2 are both acting, what is the magnitude and direction of force of friction?
- The block is at rest. What does that tell you about the max. static friction?
- When F_1 is removed, what is the friction force as a result of F_2 ? How does this frictional force compare to the max. static friction?
- Will the block start to move?